

Guia 9 – Transporte TCP

Objectivos:

- Transporte com controlo de fluxo
- Estabelecimento, controlo e terminação de sessões TCP
- Sessões TCP utilizadas comunicações FTP.

1. FTP

1. Altere o seu filtro para capturar pacotes TCP cujo endereço origem ou destino seja o endereço MAC do seu PC. Inicie uma captura e execute a seguinte sessão FTP numa janela DOS (espere cerca de 1 minuto entre cada linha de comando para mais facilmente identificar os pacotes capturados em cada passo da sessão):

```
root@gandalf:~# ftp ftp.tu-graz.ac.at
Connected to ftp.tu-graz.ac.at.
220 ProFTPD Server (ftp.tugraz.at) [129.27.3.13]
Name (ftp.tu-graz.ac.at:root): anonymous
331 Anonymous login ok, send your complete email address as your password
Password: (enter)
230-Welcome!

this is the ftp-server of graz university of technology.

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https://www.tugraz.at/ueber-diese-seite/impressum/

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https://datenschutz.tugraz.at/erklaerung/

230 Anonymous access granted, restrictions apply
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||34286|)
150 Opening ASCII mode data connection for file list
-rw-r--r-- 1 ftp ftp 266 Dec 14 2018 _datenschutzerklaerung.html
drwxr-xr-x 3 ftp ftp 35 Jun 3 2008 ftp.tu-graz.ac.at
drwxr-x--x 32 ftp ftp 4096 Jul 17 07:19 homes
-rw-r--r-- 1 ftp ftp 275 Dec 14 2018 _impressum.html
drwx-wx-wx 62 ftp ftp 4096 Oct 15 15:12 incoming
-rw-r--r-- 1 ftp ftp 275 Dec 14 2018 _legal-notice.html
drwx----- 2 ftp ftp 6 May 28 2024 lost+found
lrwxrwxrwx 1 ftp ftp 10 Jun 4 2008 mirror -> pub/mirror
drwx-wx-wx 86 ftp ftp 8192 Oct 28 09:02 outgoing
-rw-r--r-- 1 ftp ftp 266 Dec 14 2018 _privacy-policy.html
drwxr-xr-x 6 ftp ftp 81 Jul 17 07:18 pub
drwxr-xr-x 9 ftp ftp 161 Jun 26 2023 tu-graz
drwx----- 4 ftp ftp 195 Sep 29 2009 vc-graz
-rw-r--r-- 1 ftp ftp 230 Aug 30 2021 welcome.msg
```

in buscar
ficheros

```
226 Transfer complete
ftp> mget welcome.msg
mget welcome.msg [anpqy?]?
229 Entering Extended Passive Mode (|||56799|)
150 Opening BINARY mode data connection for welcome.msg (230 bytes)
100%
|*****
|*****
****| 230      189.70 KiB/s    00:00 ETA
226 Transfer complete
230 bytes received in 00:00 (4.10 KiB/s)
ftp> bye
221 Goodbye.
root@gandalf:~#
```

2. Com base na captura feita em 1, identifique e justifique:

- as ligações TCP que foram estabelecidas e qual o objetivo de cada uma
- os pacotes de estabelecimento e terminação de cada ligação TCP
- os números de porto usados em cada ligação TCP

3. Ainda com base na captura feita em 1, procure a terceira sessão TCP estabelecida entre a sua máquina e o servidor ftp.tu-graz.ac.at, e identifique:

- o pacote que estabelece a sessão
- o pacote que pede o encerramento da sessão
- o pacote que confirma o encerramento da sessão
- a relação entre cada um dos pacotes enviados pela máquina cliente e a máquina servidora

4. Com base no que observou relativamente ao encerramento da sessão explique o propósito de existirem várias mensagens TCP para encerrar uma sessão.

5. Assuma que o servidor FTP está no endereço 192.168.229.169. Inicie uma nova captura com o filtro e tente iniciar uma sessão FTP com o comando ftp 192.168.220.169. Registe e justifique os pacotes capturados.

6. Com base na relação entre os pacotes enviados e recebidos justifique a necessidade de resposta do servidor para cada mensagem TCP enviada pelo cliente, e compare o comportamento dessas máquinas com um par de máquinas (cliente e um servidor) de TFTP.

2. HTTP/HTTPS

1. Altere o seu filtro para capturar pacotes TCP cujo endereço origem ou destino seja o endereço MAC do seu PC. Inicie uma captura e execute a seguinte sessão HTTP seguinte endereço:

<https://elearning.ua.pt>

2. Com base na captura feita em 1, identifique e justifique:

- Identifique o three-way-handshake
- A ligação TCP , com os MAC address's, IP's e portos envolvidos. Crie uma tabela com a identificação de cada um dos campos.

1. FTP

1.

```
labi@labi-VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:3ae4:e597:dad5:447a prefixlen 64 scopeid 0x0<global>
    inet6 fd17:625c:f037:2:ffa5:a569:1c70:e0f8 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::6fd7:c349:a8ce:2406 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:93:d4:58 txqueuelen 1000 (Ethernet)
    RX packets 412 bytes 400086 (400.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 317 bytes 39737 (39.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

meu endereço MAC -> 08:00:27:93:d4:58

meu endereço IPv4-> 10.0.2.15

```
labi@labi-VirtualBox:~$ ftp ftp.tu-graz.ac.at
Connected to ftp.tu-graz.ac.at.
220 ProFTPD Server (ftp.tugraz.at) [129.27.3.13]
Name (ftp.tu-graz.ac.at:labi): anonymous
331 Anonymous login ok, send your complete email address as your password
Password:
230-Welcome!

this is the ftp-server of graz university of technology.

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230 Anonymous access granted, restrictions apply
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

filtro Wireshark: tcp && eth.addr == 08:00:27:93:d4:58

114	778.274845742	10.0.2.15	129.27.3.13	TCP	74	56210 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
115	778.331414927	129.27.3.13	10.0.2.15	TCP	60	21 → 56210 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
116	778.331591961	10.0.2.15	129.27.3.13	TCP	54	56210 → 21 [ACK] Seq=1 Ack=1 Win=64240 Len=0

tcp && eth.addr == 08:00:27:93:d4:58						
No.	Time	Source	Destination	Protocol	Length	Info
8	5.921669233	10.0.2.15	129.27.3.13	TCP	74	56256 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
9	5.978463697	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
10	5.978636383	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=1 Ack=1 Win=64240 Len=0
11	6.380068768	129.27.3.13	10.0.2.15	FTP	104	Response: 220 ProFTPD Server (ftp.tugraz.at) [129.27.3.13]
12	6.380159031	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=1 Ack=51 Win=64190 Len=0
13	10.360223238	10.0.2.15	129.27.3.13	FTP	70	Request: USER anonymous
14	10.360736271	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [ACK] Seq=51 Ack=17 Win=65535 Len=0
15	10.416497918	129.27.3.13	10.0.2.15	FTP	129	Response: 331 Anonymous login ok, send your complete email ad...
16	10.416568678	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=17 Ack=126 Win=64115 Len=0
17	11.737694902	10.0.2.15	129.27.3.13	FTP	61	Request: PASS
18	11.738169982	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [ACK] Seq=126 Ack=24 Win=65535 Len=0
19	11.803607370	129.27.3.13	10.0.2.15	FTP	306	Response: 230-Welcome!
20	11.803677699	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=24 Ack=378 Win=63863 Len=0
21	11.856942968	129.27.3.13	10.0.2.15	FTP	107	Response:
22	11.856994474	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=24 Ack=431 Win=63863 Len=0
23	11.857397015	10.0.2.15	129.27.3.13	FTP	60	Request: SYST
24	11.857845543	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [ACK] Seq=431 Ack=30 Win=65535 Len=0
25	11.913001365	129.27.3.13	10.0.2.15	FTP	73	Response: 215 UNIX Type: L8
26	11.954880334	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=30 Ack=450 Win=63863 Len=0

Pode ver-se o Three-Way Handshake nos primeiros 3 pacotes, criando a conexão inicial da sessão: [SYN], [SYN, ACK], [ACK].


```
labi@labi-VirtualBox:~$ ftp ftp.tu-graz.ac.at
Connected to ftp.tu-graz.ac.at.
220 ProFTPD Server (ftp.tugraz.at) [129.27.3.13]
Name (ftp.tu-graz.ac.at:labi): anonymous
331 Anonymous login ok, send your complete email address as your password
Password:
230-Welcome!
```

this is the ftp-server of graz university of technology.

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```
230 Anonymous access granted, restrictions apply
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
421 Service not available, remote server has closed connection
ftp> ls
Not connected.
```

```
labi@labi-VirtualBox:~$ ftp ftp.nluug.nl
Connected to ftp.nluug.nl.
220-Welcome to the FTP archive of
220-The Netherlands Unix Users Group (NLUUG).
220-
220-This server is located in The Netherlands, Europe.
220-If you are abroad, please find an ftp site near you.
220-Most information on this site is mirrored.
220-
220-Information about your login and any transfers you do are logged
220-If you don't like this, disconnect now.
220-
220-For statistics, see http://ftp.nluug.nl/.statistics/
220-Problems? Mail ftp-admin @ nluug.nl
220-
220-You may login as "ftp" or "anonymous".
220
Name (ftp.nluug.nl:labi): anonymous
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
500 Illegal PORT command.
ftp: bind: Address already in use
ftp> ls -la
500 Illegal PORT command.
ftp> bye
221 Goodbye.
```

```
labi@labi-VirtualBox:~$ ftp ftp.funet.fi
Connected to ftp.funet.fi.
220----- Welcome to Pure-FTPd [privsep] [TLS] -----
220-You are user number 8 of 1000 allowed.
220-Local time is now 20:51. Server port: 21.
220-Only anonymous FTP is allowed here
220-IPv6 connections are also welcome on this server.
220 You will be disconnected after 30 minutes of inactivity.
Name (ftp.funet.fi:labi): anonymous
230 Anonymous user logged in
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
500 I won't open a connection to 10.0.2.15 (only to 89.155.209.103)
ftp: bind: Address already in use
ftp> █
```

```
labi@labi-VirtualBox:~$ ftp ftp.unsw.edu.au
ftp: ftp.unsw.edu.au: Name or service not known
ftp> bye
labi@labi-VirtualBox:~$ ftp ftp.tele.ucl.ac.be
ftp: ftp.tele.ucl.ac.be: Name or service not known
ftp>
ftp>
ftp>
ftp> bye
labi@labi-VirtualBox:~$ ftp ftp.pop-mg.com.br
ftp: ftp.pop-mg.com.br: Name or service not known
ftp>
ftp> ls
Not connected.
ftp>
ftp>
ftp>
ftp>
```

```
221 Closing session.
labi@labi-VirtualBox:~$ sudo ftp test.rebex.net
[sudo] password for labi:
Connected to test.rebex.net.
220-Welcome to test.rebex.net!
    See https://test.rebex.net/ for more information and te
220 If you don't have an account, log in as 'anonymous' or
Name (test.rebex.net:labi): anonymous
331 Anonymous login OK, send your complete email address as
Password:
230 User 'anonymous' logged in.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
425 Connections to other hosts not allowed.
ftp> bye
221 Closing session.
labi@labi-VirtualBox:~$ ftp demo.wftpserver
ftp: demo.wftpserver: Name or service not known
ftp> ls
Not connected.
ftp>
ftp> bye
labi@labi-VirtualBox:~$ ftp ftp.tu-berlin.de
221 Closing session.
```

```
labi@labi-VirtualBox:~$ ftp cygwin.mirror.rafael
ftp: cygwin.mirror.rafael: Name or service not known
ftp> ls
Not connected.
ftp>
ftp> bye
labi@labi-VirtualBox:~$ ftp test.rebex.net
Connected to test.rebex.net.
220-Welcome to test.rebex.net!
    See https://test.rebex.net/ for more information and
220 If you don't have an account, log in as 'anonymous' or
Name (test.rebex.net:labi): anonymous
331 Anonymous login OK, send your complete email address
Password:
230 User 'anonymous' logged in.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
425 Connections to other hosts not allowed.
ftp> bye
221 Closing session.
```

```
Clabl@labl-VirtualBox:~$ ftp ftp.ubuntu.com
Connected to ftp.ubuntu.com.
220 FTP server (vsftpd)
Name (ftp.ubuntu.com:labl): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
500 Illegal PORT command.
ftp: bind: Address already in use
ftp> ls
500 Illegal PORT command.
ftp> bye
221 Goodbye.
labl@labl-VirtualBox:~$ ftp ftp.heanet.ie
ftp: connect to address 3.160.132.101: Connection timed out
Trying 3.160.132.120...
^Clabl@labl-VirtualBox:~$ ftp ftp.scene.org
Connected to archive.scene.org.
220 ftp.scene.org FTP server (SceneOrgFTPD-2.4.4) ready.
Name (ftp.scene.org:labl): anonymous
331 Anonymous login ok, send your complete email address as your password
Password:
230-
```

S E N D I T . ORG!

'elektronik free art' since 11 March 1996

Problems? Mail:
ftp@scene.org

```
230 Anonymous access granted, restrictions apply
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
200 PORT command successful

425 Unable to build data connection: Connection timed out
ftp>
ftp> bye
221 Goodbye.
```

2.

No.	Time	Source	Destination	Protocol	Length	Info
8	5.921669233	10.0.2.15	129.27.3.13	TCP	74	56256 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
9	5.978463697	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
10	5.978636383	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=1 Ack=1 Win=64240 Len=0
11	6.380068768	129.27.3.13	10.0.2.15	FTP	104	Response: 220 ProFTPD Server (ftp.tugraz.at) [129.27.3.13]
12	6.380159031	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=1 Ack=51 Win=64190 Len=0
13	10.360223238	10.0.2.15	129.27.3.13	FTP	70	Request: USER anonymous
14	10.360736271	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [ACK] Seq=51 Ack=17 Win=65535 Len=0
15	10.416497918	129.27.3.13	10.0.2.15	FTP	129	Response: 331 Anonymous login ok, send your complete email ad...
16	10.416568678	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=17 Ack=126 Win=64115 Len=0
17	11.737694902	10.0.2.15	129.27.3.13	FTP	61	Request: PASS
18	11.738169982	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [ACK] Seq=126 Ack=24 Win=65535 Len=0
19	11.803607370	129.27.3.13	10.0.2.15	FTP	306	Response: 230-Welcome!
20	11.803677699	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=24 Ack=378 Win=63863 Len=0
21	11.856942968	129.27.3.13	10.0.2.15	FTP	107	Response:
22	11.856994474	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=24 Ack=431 Win=63863 Len=0
23	11.857397015	10.0.2.15	129.27.3.13	FTP	60	Request: SYST
24	11.857845543	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [ACK] Seq=431 Ack=30 Win=65535 Len=0
25	11.913001365	129.27.3.13	10.0.2.15	FTP	73	Response: 215 UNIX Type: L8
26	11.954880334	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=30 Ack=450 Win=63863 Len=0

primeiros 3 pacotes são o estabelecimento da sessão aquando do comando 'ftp ftp.tu-graz.ac.at'.

51	215.478645558	10.0.2.15	129.27.3.13	FTP	78	Request: PORT 10.0.2.15,200,153
52	215.479329421	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [ACK] Seq=450 Ack=54 Win=65535 Len=0
53	215.533544107	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [RST, ACK] Seq=450 Ack=54 Win=65535 Len=0
61	234.451080198	10.0.2.15	129.27.3.13	TCP	74	39122 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=...
62	234.507510055	129.27.3.13	10.0.2.15	TCP	60	21 → 39122 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
63	234.507729996	10.0.2.15	129.27.3.13	TCP	54	39122 → 21 [ACK] Seq=1 Ack=1 Win=64240 Len=0
64	234.566112869	129.27.3.13	10.0.2.15	FTP	104	Response: 220 ProFTPD Server (ftp.tugraz.at) [129.27.3.13]
65	234.566179545	10.0.2.15	129.27.3.13	TCP	54	39122 → 21 [ACK] Seq=1 Ack=51 Win=64190 Len=0
66	239.941587121	10.0.2.15	129.27.3.13	FTP	70	Request: USER anonymous
67	239.942136640	129.27.3.13	10.0.2.15	TCP	60	21 → 39122 [ACK] Seq=51 Ack=17 Win=65535 Len=0
68	239.998310909	129.27.3.13	10.0.2.15	FTP	129	Response: 331 Anonymous login ok, send your complete email add...
69	239.998372685	10.0.2.15	129.27.3.13	TCP	54	39122 → 21 [ACK] Seq=17 Ack=126 Win=64115 Len=0
70	240.760906862	10.0.2.15	129.27.3.13	FTP	61	Request: PASS
71	240.761502065	129.27.3.13	10.0.2.15	TCP	60	21 → 39122 [ACK] Seq=126 Ack=24 Win=65535 Len=0

```

[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (24 bytes)
File Transfer Protocol (FTP)
  PORT 10.0.2.15,200,153\r\n
  Request command: PORT
  Request arg: 10.0.2.15,200,153
  Active IP address: 10.0.2.15
  Active port: 51353
[Current working directory: ]

```

Aqui pode-se verificar outro pedido TCP aquando do comando 'ls' para um porto específico, no entanto não foi possível conectar a esse porto corretamente e ver a informação disponível.

47	158.914220938	10.0.2.15	185.125.190.48	TCP	54	43010 → 80 [FIN, ACK] Seq=88 Ack=191 Win=64051 Len=0
48	158.914708401	185.125.190.48	10.0.2.15	TCP	60	80 → 43010 [ACK] Seq=191 Ack=89 Win=65535 Len=0

Aqui, é encerrada a primeira sessão TCP com um [FIN, ACK] do meu pc, e um [ACK] do servidor. (Devia ser [FIN] e [FIN, ACK]).

No.	Time	Source	Destination	Protocol	Length	Info
8	5.921669233	10.0.2.15	129.27.3.13	TCP	74	56256 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1595019025 TSecr=0 WS=128
9	5.978463697	129.27.3.13	10.0.2.15	TCP	60	21 → 56256 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
10	5.978636383	10.0.2.15	129.27.3.13	TCP	54	56256 → 21 [ACK] Seq=1 Ack=1 Win=64240 Len=0
Frame 8: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface enp0s3, id 0 Ethernet II, Src: PCSSystemtec_93:d4:58 (08:00:27:93:d4:58), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02) Internet Protocol Version 4, Src: 10.0.2.15, Dst: 129.27.3.13 Transmission Control Protocol, Src Port: 56256, Dst Port: 21, Seq: 0, Len: 0						
61	234.451080198	10.0.2.15	129.27.3.13	TCP	74	39122 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=...
62	234.507510055	129.27.3.13	10.0.2.15	TCP	60	21 → 39122 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
63	234.507729996	10.0.2.15	129.27.3.13	TCP	54	39122 → 21 [ACK] Seq=1 Ack=1 Win=64240 Len=0
Frame 61: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface enp0s3, id 0 Ethernet II, Src: PCSSystemtec_93:d4:58 (08:00:27:93:d4:58), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02) Internet Protocol Version 4, Src: 10.0.2.15, Dst: 129.27.3.13 Transmission Control Protocol, Src Port: 39122, Dst Port: 21, Seq: 0, Len: 0						

Primeira e segunda sessões usaram sempre o porto 21.

Não foi encerrada nenhuma sessão TCP do servidor FTP com [FIN] e [FIN, ACK]. No entanto, o servidor terminou a ligação com um [RST, ACK], isto pode acontecer quando um erro ocorre ou quando a conexão dá timeout após algum tempo de inatividade.

3. Só foram estabelecidas duas sessões TCP

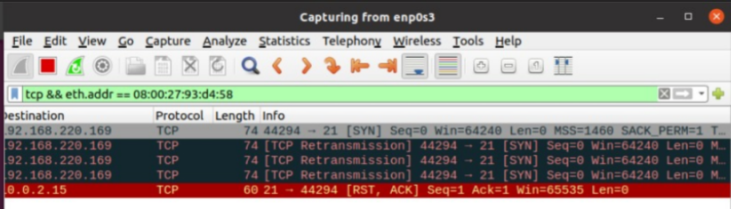
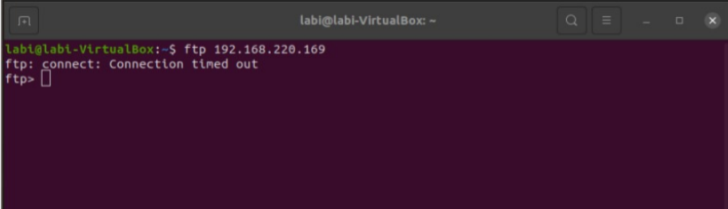
Ligação inicial:

Transferência:

Fecho da ligação:

4. Não consegui verificar, como dito anteriormente. No entanto, o propósito de existirem várias mensagens TCP para encerrar uma sessão é para o servidor ter noção de que o cliente pretende encerrar a sessão, com o [FIN]; e para o servidor efetivamente encerrar a sessão e comunicar ao cliente que a encerrou, com o [FIN, ACK].

5.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.2.15	192.168.220.169	TCP	74	44294 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3721449560 TSecr=0 WS=128
2	1.030718351	10.0.2.15	192.168.220.169	TCP	74	[TCP Retransmission] 44294 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3721450591 TSecr=0 WS=128
3	3.041008866	10.0.2.15	192.168.220.169	TCP	74	[TCP Retransmission] 44294 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3721452601 TSecr=0 WS=128
4	5.024378360	PCSSystemtec_93:d4:...	52:55:0a:00:02:02	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
5	5.024810869	52:55:0a:00:02:02	PCSSystemtec_93:d4:...	ARP	64	10.0.2.2 is at 52:55:0a:00:02:02
6	7.072576654	10.0.2.15	192.168.220.169	TCP	74	[TCP Retransmission] 44294 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3721456633 TSecr=0 WS=128
7	21.006649015	192.168.220.169	10.0.2.15	TCP	60	21 → 44294 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0

Tentou estabelecer-se a sessão com o servidor ftp em 192.168.220.169, houve retransmissão de vários [SYN], sem sucesso.

6. Como não foi possível estabelecer ligação com o servidor, não consigo justificar.

2. HTTP/HTTPS

1.

50	0.602838836	10.0.2.15	34.107.221.82	TCP	74	56922 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=471249171 TSecr=0 WS=128
51	0.604609575	fd17:625c:f037:2:5a...	2600:1901:0:38d7::	TCP	94	51248 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=439152472 TSecr=0 WS=128
52	0.615261971	2600:1901:0:38d7::	fd17:625c:f037:2:5a...	TCP	74	80 → 51248 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
53	0.617329396	fd17:625c:f037:2:5a...	2600:1901:0:38d7::	TCP	94	51256 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=439152485 TSecr=0 WS=128
54	0.617787654	2600:1901:0:38d7::	fd17:625c:f037:2:5a...	TCP	74	80 → 51256 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
55	0.619189262	192.168.86.1	10.0.2.15	DNS	203	Standard query response 0xac58 AAAA mc.prod.ads.prod.webservices.mozgcp.net SOA ns-cloud-e1.googledomains.com OPT
56	0.621562713	10.0.2.15	192.168.86.1	DNS	82	Standard query 0x6260 AAAA example.org OPT
57	0.627741898	34.107.221.82	10.0.2.15	TCP	60	80 → 56922 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
58	0.627812354	10.0.2.15	34.107.221.82	TCP	54	56922 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
59	0.627742465	192.168.86.1	10.0.2.15	DNS	146	Standard query response 0xf159 A example.org A 23.215.0.132 A 23.220.75.238
60	0.628708597	10.0.2.15	192.168.86.1	DNS	89	Standard query 0xc3ec HTTPS cloudflare-dns.com OPT
61	0.629773817	10.0.2.15	34.36.137.203	QUIC	1294	Initial, DCID=2fd16253aabd07cebb668e3326, SCID=dc1047, PKN: 0, CRYPTO
62	0.630213879	10.0.2.15	34.36.137.203	QUIC	1294	Initial, DCID=2fd16253aabd07cebb668e3326, SCID=dc1047, PKN: 1, CRYPTO
63	0.630802638	10.0.2.15	34.107.221.82	HTTP	372	GET /success.txt?ipv4 HTTP/1.1
64	0.632113746	34.107.221.82	10.0.2.15	TCP	60	80 → 56922 [ACK] Seq=1 Ack=319 Win=65535 Len=0
65	0.650808136	192.168.86.1	10.0.2.15	DNS	194	Standard query response 0x6260 AAAA example.org AAAA 2600:1408:ec00:36::173
66	0.653588639	10.0.2.15	192.168.86.1	DNS	84	Standard query 0x4ec6 A ipv4only.arpa OPT
67	0.654702994	34.36.137.203	10.0.2.15	QUIC	85	Initial, DCID=dc1047, SCID=efd16253aabd07ce, PKN: 1, ACK
68	0.655270553	10.0.2.15	192.168.86.1	DNS	84	Standard query 0x531e AAAA ipv4only.arpa OPT
69	0.657572656	34.36.137.203	10.0.2.15	QUIC	1294	Initial, DCID=dc1047, SCID=efd16253aabd07ce, PKN: 2, ACK, PADDING
70	0.657572737	34.107.221.82	10.0.2.15	HTTP	270	HTTP/1.1 200 OK (text/plain)
71	0.657608427	10.0.2.15	34.107.221.82	TCP	54	56922 → 80 [ACK] Seq=319 Ack=217 Win=64024 Len=0

50	0.602838836	10.0.2.15	34.107.221.82	TCP	74	56922 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=471249171 TSecr=0 WS=128
51	0.604609575	fd17:625c:f037:2:5a...	2600:1901:0:38d7::	TCP	94	51248 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=439152472 TSecr=0 WS=128
52	0.615261971	2600:1901:0:38d7::	fd17:625c:f037:2:5a...	TCP	74	80 → 51248 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
53	0.617329396	fd17:625c:f037:2:5a...	2600:1901:0:38d7::	TCP	94	51256 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=439152485 TSecr=0 WS=128
54	0.617787654	2600:1901:0:38d7::	fd17:625c:f037:2:5a...	TCP	74	80 → 51256 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
57	0.627741898	34.107.221.82	10.0.2.15	TCP	60	80 → 56922 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
58	0.627812354	10.0.2.15	34.107.221.82	TCP	54	56922 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
63	0.630802638	10.0.2.15	34.107.221.82	HTTP	372	GET /success.txt?ipv4 HTTP/1.1
64	0.632113746	34.107.221.82	10.0.2.15	TCP	60	80 → 56922 [ACK] Seq=1 Ack=319 Win=65535 Len=0
70	0.657572737	34.107.221.82	10.0.2.15	HTTP	270	HTTP/1.1 200 OK (text/plain)
71	0.657608427	10.0.2.15	34.107.221.82	TCP	54	56922 → 80 [ACK] Seq=319 Ack=217 Win=64024 Len=0

2.

O pacote 63 é o pedido HTTP e o pacote 70 é a resposta com a página de <https://elearning.ua.pt>

→	63	0.630802638	10.0.2.15	34.107.221.82	HTTP	372	GET /success.txt?ipv4 HTTP/1.1
←	70	0.657572737	34.107.221.82	10.0.2.15	HTTP	270	HTTP/1.1 200 OK (text/plain)

Identifique o three-way-handshake: pacotes 50, 57 e 58.

50	0.602838836	10.0.2.15	34.107.221.82	TCP	74	56922 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=471249171 TSecr=0 WS=128
51	0.604609575	fd17:625c:f037:2:5a...	2600:1901:0:38d7::	TCP	94	51248 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=439152472 TSecr=0 WS=128
52	0.615261971	2600:1901:0:38d7::	fd17:625c:f037:2:5a...	TCP	74	80 → 51248 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
53	0.617329396	fd17:625c:f037:2:5a...	2600:1901:0:38d7::	TCP	94	51256 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=439152485 TSecr=0 WS=128
54	0.617787654	2600:1901:0:38d7::	fd17:625c:f037:2:5a...	TCP	74	80 → 51256 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
55	0.619189262	192.168.86.1	10.0.2.15	DNS	203	Standard query response 0xac58 AAAA mc.prod.ads.prod.webservices.mozgcp.net SOA ns-cloud-e1.googledomains.com OPT
56	0.621562713	10.0.2.15	192.168.86.1	DNS	82	Standard query 0x6260 AAAA example.org OPT
57	0.627741898	34.107.221.82	10.0.2.15	TCP	60	80 → 56922 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
58	0.627812354	10.0.2.15	34.107.221.82	TCP	54	56922 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0

A ligação TCP:

50	0.602838836	10.0.2.15	34.107.221.82	TCP	74	56922 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=471249171 TSecr=0 WS=128
51	0.604609575	fd17:625c:f037:2:5a...	2600:1901:0:38d7::	TCP	94	51248 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=439152472 TSecr=0 WS=128
52	0.615261971	2600:1901:0:38d7::	fd17:625c:f037:2:5a...	TCP	74	80 → 51248 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
53	0.617329396	fd17:625c:f037:2:5a...	2600:1901:0:38d7::	TCP	94	51256 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=439152485 TSecr=0 WS=128
54	0.617787654	2600:1901:0:38d7::	fd17:625c:f037:2:5a...	TCP	74	80 → 51256 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0
57	0.627741898	34.107.221.82	10.0.2.15	TCP	60	80 → 56922 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
58	0.627812354	10.0.2.15	34.107.221.82	TCP	54	56922 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0

▶	Frame 50: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface enp0s3, id 0
▶	Ethernet II, Src: PCSSystemtec_93:d4:58 (08:00:27:93:d4:58), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02)
▶	Internet Protocol Version 4, Src: 10.0.2.15, Dst: 34.107.221.82
▶	Transmission Control Protocol, Src Port: 56922, Dst Port: 80, Seq: 0, Len: 0
	Source Port: 56922
	Destination Port: 80

MAC address's: meu endereço -> 08:00:27:93:d4:58
endereço de destino -> 52:55:0a:00:02:02

IP's: meu endereço -> 10.0.2.15
endereço de destino -> 34.107.221.82

Portos envolvidos:

No three-way handshake, o meu porto é o 56922 e o porto de destino para onde é feito o pedido HTTP é o porto 80.